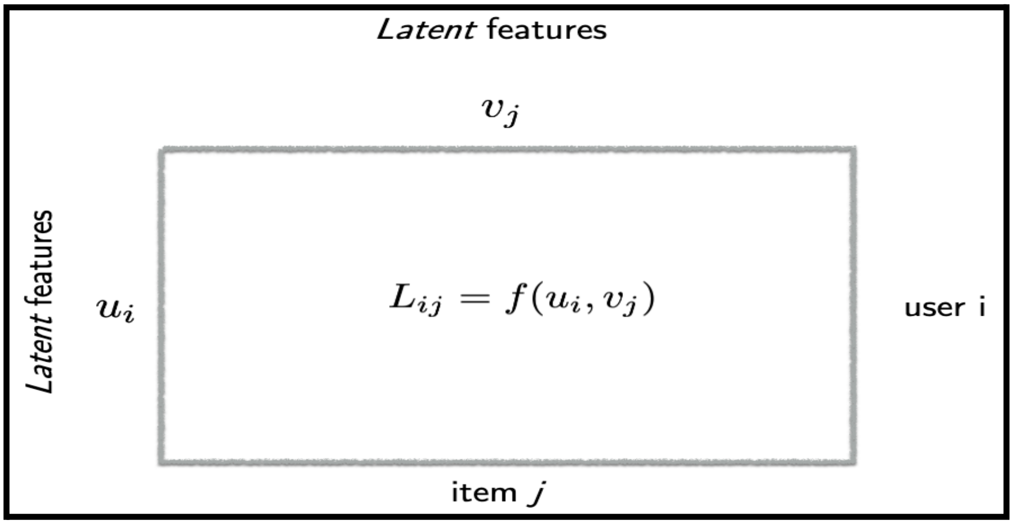
1.8 Recommendation Systems - Singular Value Threshold

Prediction problem: Complete the matrix



To estimate L_{ij} for any user i and item j using SVD, we extract the latent features that explain the behavior of both users and items, i.e., we try to find u_i and v_j (where u_i and v_j represent latent features of the rows and columns, respectively).

Let's suppose there are some latent features U and V for the users and items. The features have many missing values, so the purpose of SVD is to estimate the matrix by filling null values with 0 and applying the following formula to estimate the feature matrix.

$$A = USV^T$$

Where, U and V are the Left Singular matrix and Right Singular matrix, respectively and S is a diagonal matrix.

$$\underset{m \times n}{\overset{X}{\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & \\ \vdots & \vdots & \ddots & \\ x_{m1} & & & x_{mn} \end{pmatrix}}} = \underset{m \times r}{\overset{U}{\begin{pmatrix} u_{11} & \cdots & u_{1r} \\ \vdots & \ddots & \\ u_{m1} & & u_{mr} \end{pmatrix}}} \underset{r \times r}{\overset{S}{\begin{pmatrix} s_{11} & 0 & \cdots \\ 0 & \ddots & \\ \vdots & & s_{rr} \end{pmatrix}}} \underset{r \times n}{\overset{V^{\top}}{\begin{pmatrix} v_{11} & \cdots & v_{1n} \\ \vdots & \ddots & \\ v_{r1} & & v_{rn} \end{pmatrix}}}$$

An example:

$$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$$

Now, any matrix obeys singular value decomposition if rank(X) = r

To find the SVD of L:

Step 1: Fill missing values in L by 0. (In case of causal dependency a better way is to fill the missing values by taking the average of the row and column values)

Step 2: Compute SVD

$$L_{ij} = \sum_{k=1}^{min(M,N)} s_k u_{ik} v_{jk}$$

Step 3: Apply truncated SVD by keeping the top r components (+ normalize)

$$\widehat{L}_{ij} = \frac{1}{\hat{p}} \sum_{k=1}^{r} s_k u_{ik} v_{jk} \text{ for all i,j}$$

Where $\hat{p}$ is the fraction of observed entries.

Question: How do we choose r from step 3?

Ans: We first sort the singular values which we obtained in step 1 in descending order and plot the values in a graph. We consider r as the value where the knee point appears in the plot. In case the graph is smooth, then it means that this algorithm doesn't work properly for the dataset

Example of SVD:

Question: We were given that r=1, s_1=1 and we know the values of the first-row and first column. Find the values of the matrix.

The SVD formula is

$$\widehat{L_{ij}} = \frac{1}{\hat{p}} \sum_{k=1}^{r} s_k u_{ik} v_{jk}$$

Given r=1 and s_1=1 then the equation becomes

$$L_{ij} = u_i v_j$$

Now, we adjust the formula as (multiplying and dividing by $u_1 v_1$)

$$L_{ij} = \frac{\left(u_i v_1\right) * \left(u_1 v_j\right)}{\left(u_1 v_1\right)}$$

We know that $\left(u_i v_1\right)$ is L_{i1} and $\left(u_1 v_j\right)$ is L_{1j}

$$L_{ij} = \frac{\left(L_{i1}\right) * \left(L_{1j}\right)}{\left(L_{11}\right)}$$

As given in the question, we know the values of the first row and column, i.e., we know the values of L_{i1} and the values of the first column L_{1j}. Now, we can recover all the values of the matrix by changing values for i and j.

- From the above matrix with $m + n - 1$ observations, we can recover $m \times n$ values of the matrix.
- We can generalize the above statement as if we have rank r matrix we roughly need to know $(m + n) \times r$ entries to recover the entire $m \times n$ matrix.

Optimization of SVD:

Singular value decomposition: An optimization perspective

In case we only observe a few observations of the matrix which are noisy we can still find the U and V matrices such that the sum of the squared error on the observed entries is minimized.

Mathematically, the optimization problem of finding U and V can be written as:

$$\text{minimize} \sum_{(i,j)\,\epsilon\, obs} \left(L_{ij} - \sum_{k=1}^{r} u_{ik} v_{jk}\right)^2$$

over

$$u_{ik} \epsilon R,\ 1 \le i \le N, 1 \le k \le r$$

$$v_{jk} \epsilon R,\ 1 \le j \le M, 1 \le k \le r$$

We can solve the above optimization problem by making it into 2 subproblems:

Step 1: Fix V (items) and solve for U (items) by minimization of the cost function, we can think of it as a linear regression problem on i keeping j as fixed.

$$\text{i.e., minimize} \sum_{i \epsilon obs} \left(L_{ij} - \sum_{k=1}^{r} (u_{ik}) v_{jk}\right)^2$$

as we know L_{ij} and u_{ik} let's think of v_{jk} as $\beta's$ and the equation becomes of the form:

$$\text{minimize} \sum_{i \epsilon obs} \left(Y_i - x_i \times \beta\right)^2$$

Step 2: Similarly, now Fix U (users) and solve for V (items) by minimization of the cost function, we can think of it as a linear regression problem on j keeping i as constant.

i.e., minimize $\sum_{j \epsilon obs} \left(L_{ij} - \sum_{k=1}^{r} u_{ik}(v_{jk}) \right)^2$

minimize $\sum_{j \epsilon obs} \left(Y_j - x_j \times \beta \right)^2$

- After finding the U and V

  For any i, j, produce estimate as follows:

$$\widehat{L_{ij}} = \sum_{k=1}^{r} u_{ik} v_{jk}.$$

Note: The above algorithm uses only observed entries and does not require filling missing values as in for SVD.